

Author-Review Items from the Proofreading Pass

July 2, 2026

This file lists only issues that still need an author decision or follow-up edit. Local fixes already applied in LyX are intentionally omitted.

1 Items Needing Author Review

1.1 Generalized gradient descent: specify $0 < \alpha \leq 1$

Location: `gradient_descent.lyx`, generalized gradient descent / α -approximation.

Issue. The text says the approximation factor is $\alpha > 0$. The proof constructs

$$z = (1 - \alpha)x^{(k)} + \alpha\hat{x}$$

and uses convexity-type inequalities as if z is a convex combination. That requires $0 < \alpha \leq 1$. The convergence rate $(1 - \alpha)^k$ also only has its usual meaning in this range.

Action needed. Change the definition and theorem assumptions from “ $\alpha > 0$ ” to “ $0 < \alpha \leq 1$ ”, for example:

For $0 < \alpha \leq 1$, we say $g + h$ is an α -approximation to f .

Then update the theorem statement to assume the algorithm can find such an α -approximation for some $0 < \alpha \leq 1$.

1.2 Center-of-gravity tail bound proof: justify the $|t|$ statement

Location: `cutting_plane.lyx`, proof of the lower tail bound after Grunbaum.

Issue. The theorem states a bound of the form

$$\mathbb{P}(x^\top \theta \geq t) \geq \frac{1}{e} - |t|.$$

The proof as written ends with a statement for $t \geq 0$ and uses $P(0) \geq 1/e$, but it does not explicitly justify the upper-tail probability at zero or the case $t < 0$. The argument should be symmetric or should invoke the same reasoning for $-x$ and then handle negative t separately.

Action needed. After proving $P(0) \geq 1/e$, add a sentence explaining that applying the same argument to $-x$ gives the corresponding upper-tail mass at zero. Then split into cases:

$$t \geq 0 : \quad P(x^\top \theta \geq t) \geq P(x^\top \theta \geq 0) - t \geq 1/e - t,$$

and

$$t < 0 : \quad P(x^\top \theta \geq t) \geq P(x^\top \theta \geq 0) \geq 1/e \geq 1/e - |t|.$$

1.3 Cutting-plane volume proof: final body is not necessarily a cuboid

Location: `cutting_plane_volume.lyx`, theorem proving `CuttingPlaneVolume` computes volume.

Issue. The proof says that when the algorithm stops, the remaining set “must be an axis-parallel cuboid” with side lengths in $[r/(n+1), r]$. A convex body whose coordinate widths are all small is contained in an axis-parallel box, but it need not equal that box. Therefore the product of coordinate widths is not automatically the volume of the remaining body. This matters because the algorithm’s return expression multiplies the estimated ratio V_{last} by a product of widths.

Action needed. Decide which theorem is intended:

1. If this is meant as an exact deterministic reduction to centroid computation, the algorithm needs an invariant or stopping condition that makes the final set exactly a box/cuboid, or it needs to continue with a procedure that computes the final small body’s volume exactly.
2. If this is meant as an approximation, change the theorem statement from “correctly computes the volume” to an approximation guarantee, and bound the error introduced by replacing the final body’s volume by the product of coordinate widths.
3. If the product of widths is only meant as an upper or lower bound, state that explicitly and adjust the proof accordingly.

1.4 Reduction-rate prose: check the displayed rates in the narrative

Location: `reduction.lyx`, reduction summary around the complexity table.

Issue. A reviewer noted that the prose says the number of iterations of gradient descent is GR/ϵ , while the table and standard nonsmooth Lipschitz rate use

$$\left(\frac{GR}{\epsilon}\right)^2.$$

Nearby prose also lists terms such as GR/k , $G^2/(\mu k)$, and LR^2/k , whereas the usual rates being discussed appear to be $GR/\sqrt{k}$, $G^2/(\mu k)$, and LR^2/k^2 , respectively.

Action needed. Re-read this paragraph against Table 5.1 and the intended oracle models. If the paragraph is discussing function-error after k iterations, replace the rates with the matching table rates.

1.5 Annealing-volume schedule: exact value of m

Location: `annealing_volume.lyx`, Algorithm 34 and proof of the volume theorem.

Issue. The algorithm currently sets

$$m = \left\lceil \sqrt{n} \ln(4R^2 n / \epsilon) \right\rceil, \quad a_i = \frac{2n}{(1 + 1/\sqrt{n})^i}.$$

Later the proof uses $a_m \leq \epsilon/(4R^2)$. This follows directly if

$$(1 + 1/\sqrt{n})^m \geq \frac{8R^2 n}{\epsilon},$$

or equivalently

$$m \geq \log_{1+1/\sqrt{n}}(8R^2 n / \epsilon).$$

The present formula with $\sqrt{n} \ln(4R^2 n/\epsilon)$ is asymptotically similar but does not visibly imply the exact constant used later without an additional inequality check.

Action needed. Either set

$$m = \left\lceil \log_{1+1/\sqrt{n}}(8R^2 n/\epsilon) \right\rceil$$

in the algorithm, or keep the asymptotic-looking expression and add a line proving it implies $a_m \leq \epsilon/(4R^2)$ with the stated constants.

1.6 Annealing-volume variance proof: logconcavity should be for $a^n F(a)$

Location: `annealing_volume.lyx`, proof of the ratio variance lemma.

Issue. The proof says that F is logconcave in a . The referenced lemma appears to prove logconcavity of $a^n F(a)$, and the displayed bound uses the extra a^n factors exactly as if

$$Z(a) = a^n F(a)$$

is the logconcave function.

Action needed. Replace the sentence with:

By Lemma 11.10, $Z(a) = a^n F(a)$ is logconcave in a .

Then derive the variance bound using logconcavity of Z , not of F alone.

1.7 Hit-and-Run section: empty exercise wrapper around a figure

Location: `har_dikin.lyx`, just before the Hit-and-Run trajectory figure.

Issue. The PDF shows an “Exercise.” label immediately before the figure, but there is no exercise text attached to it. This looks like an accidental empty exercise environment.

Action needed. Either remove the empty exercise wrapper, or add the intended exercise statement before the figure.

1.8 Dikin walk algorithm: specify rejection behavior

Location: `har_dikin.lyx`, Dikin walk algorithm.

Issue. The algorithm samples a proposal z and says that if $z \in K$, set

$$x^{(i+1)} = z$$

with a Metropolis probability. It does not explicitly say what happens when $z \notin K$, nor what happens when the Metropolis proposal is rejected. The intended Markov chain presumably stays at $x^{(i)}$ in both cases.

Action needed. Add the missing rejection cases, for example:

$$\text{otherwise set } x^{(i+1)} = x^{(i)}.$$

This should cover both $z \notin K$ and failure of the acceptance test.